

DD_SIP-04 - Bundle Launch

Version: 1.0 Status: Draft for review Bundle: 09_catalogs Audience: product catalog developers, commercial developers, subscription developers, billing developers, frontend developers, QA Related DDs:

- DD_SIP-01 - Package Management
- DD_PLM-CFG-01 - Service Catalog
- DD_PLM-CFG-04 - Discount Catalog
- DD_SIP-03 - Discount Assignment
- DD_SIP-05 - Promotional Campaigns
- DD_SUB-WF-MIGRATION-01 - Subscription Migration
- DD_SUB-WF-UPGRADE-01 - Subscription Upgrade
- DD_SUB-WF-DOWNGRADE-01 - Subscription Downgrade
- DD_BIL-01 - Charging Engine Core
- DD_FUL-02 - Order Capture
- DD_EM-CFG-04 - Approval Workflow Catalog
- FOUNDATION_DROOLS - Rules Engine Foundation
- FE-APP-01 - BSS Backoffice Web App
- FE-APP-02 - Outdoor Sales Franchise App

1. Purpose

SIP-04 defines how SOPHIX launches commercial bundles and controls bundle-to-bundle migration paths.

A bundle is a commercial offer made of one or more existing package references, optional one-off charges, optional discount assignment rules, and channel/region/franchise availability. Examples:

- Fiber Internet + IPTV starter bundle;
- Internet + Voice bundle for residential customers;
- franchise acquisition bundle with free installation discount;
- business bundle with static IP, higher SLA, and recurring add-on;
- migration bundle replacing an older offer with a newer commercial offer.

SIP-04 does not define low-level services. Services are owned by PLM-CFG-01. Packages are owned by SIP-01. Discounts are owned by PLM-CFG-04/SIP-03. SIP-04 coordinates these catalog items into a launchable commercial bundle and publishes rules that order capture, sales, subscription workflows, and billing can consume.

2. Ownership Boundary

Concern	Owner
Service definition	PLM-CFG-01
Package definition and package versions	SIP-01
Bundle launch, availability, commercial composition, migration rules	SIP-04
Discount definition	PLM-CFG-04
Discount assignment when bundle is ordered	SIP-03
Promotional campaign governance	SIP-05
Configurable launch approval process	EM-CFG-04
Customer order capture	FUL-02
Active subscription migration/upgrade/downgrade execution	SUB workflows

Concern	Owner
Billing-time charge/discount calculation	BIL-01 / DIS-OP-01
Configurable launch/migration eligibility decisions	Drools rules engine where policy is operator-variable

Rule: SIP-04 publishes bundle catalog and migration rules. It must not directly change existing subscriptions or create invoices. SUB/FUL/BIL modules execute customer-facing operations using SIP-04 rules.

Approval rule: SIP-04 does not define approval steps locally. Launch approval should call EM-CFG-04 for BUNDLE_LAUNCH_APPROVAL or similar action codes. Camunda is used only when the selected EM-CFG-04 policy is BPMN-backed.

3. When To Use This DD

Use SIP-04 when:

- product team needs to launch a multi-package commercial offer;
- sales team needs one selectable offer made of several packages;
- existing customers need migration from old bundle to new bundle;
- bundle availability differs by franchise, region, technology, channel, or customer segment;
- launch requires pre-checks before becoming sellable.

Do not use SIP-04 for:

Scenario	Use
Creating a single package and its price/version	SIP-01
Creating a discount definition	PLM-CFG-04
Assigning a discount to a customer/order	SIP-03
Running a promotion with dates, cohorts, and channels	SIP-05
Moving one customer's subscription	SUB-WF-MIGRATION/UPGRADE/DOWNGRADE

4. High Level Flow

```

flowchart TD
    A["Product manager drafts bundle"]
    B["Select package components"]
    C["Attach optional discounts/charges"]
    D["Configure availability and channels"]
    E["Validate package, price, tech, franchise, billing rules"]
    F{"Approval required?"}
    G["Approval review"]
    H["Schedule launch"]
    I["Activate bundle"]
    J["Publish BundleLaunched event"]
    K["Sales/FUL/SUB/BIL caches refresh"]
    L["Customers order or migrate"]

    A --> B --> C --> D --> E --> F
    F -->|Yes| G --> H
    F -->|No| H
    H --> I --> J --> K --> L

```

5. Bundle Lifecycle

Status	Meaning	Allowed next statuses
DRAFT	Bundle is being prepared. Not visible to sales/order capture.	READY_FOR_REVIEW, CANCELLED
READY_FOR_REVIEW	Validation passed and waiting for approval/release.	APPROVED, DRAFT, REJECTED
APPROVED	Launch approved but not sellable until launch date.	ACTIVE, CANCELLED

Status	Meaning	Allowed next statuses
ACTIVE	Bundle can be selected by allowed channels/regions.	SUSPENDED, RETIRED
SUSPENDED	Temporarily hidden for new orders/migrations. Existing subscriptions unaffected.	ACTIVE, RETIRED
RETIRED	No new sales/migrations. Kept for history.	none
REJECTED	Review rejected.	DRAFT
CANCELLED	Launch cancelled before activation.	none

Existing customers do not automatically move when a bundle is retired. Migration requires explicit SUB workflow execution.

6. Tables

6.1 commercial_bundle

Stores the bundle master record.

Use this table when product/commercial teams create a named offer that combines packages and optional commercial rules.

Column	Type	Purpose	Sample
bundle_id	UUID PK	Bundle id.	bun-001
operator_code	TEXT	Operator/tenant.	WIK
bundle_code	TEXT	Stable unique code.	FIBER_TV_STARTER
display_name	TEXT	Customer/agent visible name.	Fiber + TV Starter
description	TEXT NULL	Internal/customer description.	100M fiber with starter IPTV package.
status	TEXT	Lifecycle status.	ACTIVE
bundle_type	TEXT	ACQUISITION, RETENTION, MIGRATION, BUSINESS, STAFF, GENERAL.	ACQUISITION
currency_code	TEXT	Pricing currency.	KES
launch_date	DATE	First sellable date.	2026-07-01
retire_date	DATE NULL	Planned retirement.	2026-12-31
created_by_user_id	TEXT	Creator.	usr-product-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-10T09:00:00Z
updated_at	TIMESTAMP	Last update.	2026-06-10T09:10:00Z

Recommended indexes:

- Unique (operator_code, bundle_code).
- (operator_code, status, launch_date).

6.2 commercial_bundle_component

Stores packages included in a bundle.

Use this table to define what the customer receives when they buy the bundle. Components reference SIP-01 packages or package versions.

Column	Type	Purpose	Sample
component_id	UUID PK	Component id.	bcomp-001
bundle_id	UUID	Bundle id.	bun-001
package_ref	TEXT	SIP-01 package code/id.	pkg-fiber-100m
package_version_id	TEXT NULL	Optional pinned version.	pkv-100m-202607
component_role	TEXT	PRIMARY, ADDON, OPTIONAL, PROMOTIONAL.	PRIMARY
quantity	INT	Quantity/units.	1

Column	Type	Purpose	Sample
mandatory	BOOLEAN	Must be included in order.	true
display_order	INT	UI display order.	10
metadata_json	JSONB	Extra details.	{"speed": "100M"}

Recommended indexes:

- (bundle_id, display_order).
- (package_ref).

6.3 commercial_bundle_availability

Stores where and how a bundle can be sold.

Use this table for country/region/franchise/technology/channel gating.

Column	Type	Purpose	Sample
availability_id	UUID PK	Availability id.	bav-001
bundle_id	UUID	Bundle id.	bun-001
country_code	TEXT	Country.	KE
region_code	TEXT NULL	Commercial/tech region.	NRB
franchise_id	TEXT NULL	Franchise scope.	fr_01J_NRB_002
tech_region_code	TEXT NULL	Network technology region.	GPON-NRB-01
channel_code	TEXT	BACKOFFICE, SALES_APP, SELF_CARE, USSD, PARTNER_API.	SALES_APP
status	TEXT	ACTIVE, SUSPENDED, RETIRED.	ACTIVE
effective_from	DATE	Start date.	2026-07-01
effective_to	DATE NULL	End date.	null

Recommended indexes:

- (bundle_id, status).
- (country_code, region_code, channel_code, status).
- (franchise_id, status).

6.4 commercial_bundle_discount_rule

Stores optional discount assignment rules attached to a bundle.

Use this table when ordering the bundle should create a SIP-03 discount assignment, such as free installation or three months discounted recurring charge.

Column	Type	Purpose	Sample
discount_rule_id	UUID PK	Rule id.	bdr-001
bundle_id	UUID	Bundle id.	bun-001
discount_code	TEXT	PLM-CFG-04 discount code.	ACQ_FREE_INSTALL
assignment_scope_type	TEXT	Scope passed to SIP-03.	ORDER
validity_mode	TEXT	ORDER_ONLY, FIXED_WINDOW, DURATION_FROM_ACTIVATION.	ORDER_ONLY
duration_days	INT NULL	Duration if dynamic.	90
priority	INT	Runtime priority.	100
stacking_group_code	TEXT NULL	Stacking group.	INSTALLATION
status	TEXT	ACTIVE, SUSPENDED, RETIRED.	ACTIVE

Recommended indexes:

- (bundle_id, status).
- (discount_code).

6.5 commercial_bundle_migration_rule

Stores allowed bundle-to-bundle movement rules.

Use this table when existing customers may move from one bundle to another and the operator wants controlled upgrade/downgrade/migration paths.

Column	Type	Purpose	Sample
migration_rule_id	UUID PK	Rule id.	bmr-001
operator_code	TEXT	Operator/tenant.	WIK
source_bundle_id	UUID	Current bundle.	bun-old-001
target_bundle_id	UUID	Target bundle.	bun-001
movement_type	TEXT	UPGRADE, DOWNGRADE, MIGRATION, RETENTION, FORCED_RETIREMENT.	MIGRATION
allowed_channel_json	JSONB	Allowed channels.	["BACKOFFICE", "SELF_CARE"]
requires_customer_consent	BOOLEAN	Customer must confirm.	true
requires_wo	BOOLEAN	Field work required.	false
fee_policy_code	TEXT	Fee policy.	NO_FEE
effective_from	DATE	Start date.	2026-07-01
effective_to	DATE NULL	End date.	null
status	TEXT	ACTIVE, SUSPENDED, RETIRED.	ACTIVE

Recommended indexes:

- (source_bundle_id, target_bundle_id, status).
- (operator_code, movement_type, status).

6.6 commercial_bundle_launch_check

Stores launch validation findings.

Use this table to make launch readiness auditable. Failed checks explain why a bundle cannot be activated.

Column	Type	Purpose	Sample
check_id	UUID PK	Check id.	blc-001
bundle_id	UUID	Bundle id.	bun-001
check_code	TEXT	Validation check.	PACKAGE_ACTIVE
check_status	TEXT	PASS, WARN, FAIL.	PASS
message	TEXT	Human-readable result.	All package refs are active.
source_ref_json	JSONB	Supporting refs.	{"packageRef": "pkg-fiber-100m"}
checked_at	TIMESTAMP	Check timestamp.	2026-06-10T09:15:00Z

Recommended indexes:

- (bundle_id, check_status).

7. APIs

7.1 Create Bundle

```
POST /api/commercial-bundles
Idempotency-Key: bundle-fiber-tv-starter-v1
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "bundleCode": "FIBER_TV_STARTER",
  "displayName": "Fiber + TV Starter",
  "bundleType": "ACQUISITION",
  "currencyCode": "KES",
  "launchDate": "2026-07-01",
```

```

    "components": [
      { "packageRef": "pkg-fiber-100m", "componentRole": "PRIMARY", "mandatory": true },
      { "packageRef": "pkg-iptv-starter", "componentRole": "ADDON", "mandatory": true }
    ]
  }
}

```

7.2 Validate Bundle

```
POST /api/commercial-bundles/{bundle_id}/validate
```

Validation checks package existence/status, version availability, technology compatibility, target regions/franchises, discount refs, and billing event compatibility.

7.3 Submit For Review

```
POST /api/commercial-bundles/{bundle_id}/submit-review
```

Allowed only when validation has no FAIL.

7.4 Approve Launch

```

POST /api/commercial-bundles/{bundle_id}/approve
Content-Type: application/json

{
  "decisionComment": "Approved for Nairobi sales launch."
}

```

7.5 Activate Bundle

```
POST /api/commercial-bundles/{bundle_id}/activate
```

Activation can happen immediately on/after `launch_date`, or by scheduled worker.

7.6 Get Available Bundles

Used by Backoffice, Sales App, Self-care, and order capture.

```
GET /api/commercial-bundles/available?operatorCode=WIK&channelCode=SALES_APP&franchiseId=fr_01J_NRB_002&homepassId=HP-001
```

Response:

```

{
  "bundles": [
    {
      "bundleId": "bun-001",
      "bundleCode": "FIBER_TV_STARTER",
      "displayName": "Fiber + TV Starter",
      "components": [
        { "packageRef": "pkg-fiber-100m", "displayName": "Fiber 100M" },
        { "packageRef": "pkg-iptv-starter", "displayName": "IPTV Starter" }
      ],
      "discountRules": [
        { "discountCode": "ACQ_FREE_INSTALL", "assignmentScopeType": "ORDER" }
      ]
    }
  ]
}

```

7.7 Preview Bundle Migration

```

POST /api/commercial-bundles/migration-preview
Content-Type: application/json

```

```

{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "sourceBundleCode": "OLD_FIBER_TV",
  "targetBundleCode": "FIBER_TV_STARTER",
  "channelCode": "BACKOFFICE"
}

```

Response:

```

{
  "allowed": true,
  "movementType": "MIGRATION",
  "requiresCustomerConsent": true,
  "requiresWo": false,
  "feePolicyCode": "NO_FEE",
}

```

```

    "owningWorkflow": "SUB-WF-MIGRATION-01"
  }

```

SIP-04 only previews and authorizes the path. SUB workflow executes the customer migration.

8. Events

Event	Topic	When emitted
CommercialBundleCreated	sophix.commercial.bundle.created	Bundle draft created.
CommercialBundleValidated	sophix.commercial.bundle.validated	Validation finished.
CommercialBundleApproved	sophix.commercial.bundle.approved	Launch approved.
CommercialBundleActivated	sophix.commercial.bundle.activated	Bundle becomes sellable.
CommercialBundleSuspended	sophix.commercial.bundle.suspended	Bundle hidden from new sales.
CommercialBundleRetired	sophix.commercial.bundle.retired	Bundle no longer sellable.
BundleMigrationRuleUpdated	sophix.commercial.bundle.migrationrule updated	Migration rule changed.

9. Business Rules

Rule	Description	Reason
R-SIP-BUN-01	Bundle code must be unique per operator.	Stable external reference.
R-SIP-BUN-02	Bundle must have at least one mandatory package component.	Avoid empty offers.
R-SIP-BUN-03	Package refs must exist and be active or launch-ready.	Prevent unsellable offers.
R-SIP-BUN-04	Bundle availability must intersect component package availability.	Prevent sales in unsupported regions.
R-SIP-BUN-05	Discount rules must reference active discount catalog items.	Prevent invalid order discounts.
R-SIP-BUN-06	ACTIVE bundle cannot change component composition; create a new bundle/version instead.	Preserve order/subscription history.
R-SIP-BUN-07	Retiring a bundle blocks new orders but does not cancel existing subscriptions.	Customer continuity.
R-SIP-BUN-08	Bundle migration must use an active migration rule.	Controlled movement.
R-SIP-BUN-09	Customer migration execution must be delegated to SUB workflow.	Ownership boundary.
R-SIP-BUN-10	Sales/self-care can show only available bundles for channel, region, franchise, customer, and date.	Avoid invalid promises.

10. Rule Engine Guidance

Use Drools for bundle decisions that are operator-variable or combine several facts:

Rule file	Purpose
bundle-launch-validation.drl	Launch readiness beyond fixed DB/API checks, such as minimum package mix or channel restrictions.
bundle-availability.drl	Dynamic availability by channel, franchise, region, customer segment, or date.
bundle-migration-eligibility.drl	Whether a customer can move from one bundle to another and which SUB workflow owns execution.

Fixed checks such as bundle code uniqueness, package existence, and date parsing remain normal application validation.

Example rule:

```

rule "Self-care migration requires consent"
when
    $m : BundleMigrationFact(channelCode == "SELF_CARE", movementType == "MIGRATION")
    $out : BundleMigrationDecision()

```

```

then
  $out.setRequiresCustomerConsent(true);
  $out.setOwningWorkflow("SUB-WF-MIGRATION-01");
end

```

11. Frontend Usage

11.1 FE-APP-01 Backoffice

Backoffice must support:

- create/edit bundle drafts;
- add package components;
- configure availability and discount rules;
- run validation checks;
- submit/approve launch;
- activate/suspend/retire;
- manage migration rules;
- preview migration from a customer/subscription context.

11.2 FE-APP-02 Sales App

Sales app uses read-only availability APIs:

- list available bundles for lead/HomePass/franchise/channel;
- show bundle components and total offer summary;
- show attached promotional terms;
- submit selected bundle into lead/order conversion payload.

Sales app must not define bundle rules locally. It displays bundle results returned by SIP-04/SIP-05 APIs.

12. Infra Recommendation

For a medium budget deployment, run SIP-04 inside the existing commercial/catalog API deployment. Do not create a separate pod initially.

Component	Recommendation
API	Existing catalog/commercial API pods expose <code>/api/commercial-bundles</code> .
DB	Same PostgreSQL cluster/schema family as SIP package/commercial catalog.
Workers	Existing worker pool handles scheduled activation, retirement, validation refresh.
Events	Kafka/MSK topics under <code>sophix.commercial.bundle.*</code> .
Cache	Read model/cache for available bundles by channel/franchise/region.

AWS deployment: use existing ECS/EKS service and RDS cluster. Add alarms for activation worker failure, validation failures on approved bundles, stale availability cache, and migration preview error rate.

13. Testing Requirements

13.1 Unit Tests

- Bundle code uniqueness.
- Package component validation.
- Availability intersection with package region/franchise.

- Discount rule validation.
- Lifecycle transition rules.
- Migration rule lookup.
- Drools availability and migration eligibility decisions.

13.2 Integration Tests

- Create bundle using SIP-01 package refs.
- Validate and activate bundle.
- Available bundles API filters by channel/franchise/date.
- Bundle discount rules are passed to order/assignment flow.
- Migration preview returns SUB workflow owner.

13.3 End To End Tests

- Product launches Fiber + TV bundle; sales app sees it for eligible HomePass.
- FUL order created with bundle ref and package components.
- Bundle retirement blocks new orders but existing subscription remains active.
- Customer migration from old bundle to new bundle uses SUB migration workflow.

14. Definition Of Done

- Bundle tables and APIs exist.
- Bundle lifecycle is implemented and auditable.
- Bundle validation checks package, discount, availability, and billing compatibility.
- Available bundle API is usable by Backoffice, Sales App, order capture, and self-care.
- Migration rules exist and delegate execution to SUB workflows.
- Drools rule files exist for configurable launch, availability, and migration decisions where operator policy requires them.
- Events refresh downstream caches.
- Tests cover launch, availability, retirement, and migration preview.